# Quantizable Models for M-theory

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**Abstract**

The hidden exceptional symmetries of eleven-dimensional supergravity that emerge upon compactification on tori are believed to provide important hints to the formulation of M-theory as it results from the quantization of some yet unknown fundamental objects generalizing strings. In 1985, former Aggie Michael Duff conjectured that this theory should have the full exceptional symmetry present as a spacetime symmetry already in eleven dimensions. In this talk I will present models with the property that their canonical quantization gives rise to current algebras on extended objects with these exceptional global symmetries. (These are known for ranks up to and including seven.) In particular, truncation of the bracket of two vector field currents to the massless level reproduces the "exceptional Courant bracket" of the exceptional/generalized geometry. Such geometries come with a "section condition" on their momenta which arises in these models from a Virasoro constraint.

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# 1 Review

## 1.1 S-theory

**Pre-1984:** There is a unique 11D supergravity and there are five 10D supergravities of types I, IIA/B, HE/O.

**Green & Schwarz (1984) [5]:** 10D supergravities "$S(X)$" on spacetime $X$ originate from quantization of fundamental strings "$\mathbf{S}(X)$" with target $X$.

Moduli space of Type-II supergravity $S(X)$ on $X \to X \times T^n$:

$$\frac{O(n,n)}{O(n) \times O(n)}$$

due to the symmetry between momentum modes (one copy of $n$) and winding modes (the other copy of $n$).

**Narain (1986) [6]:** $O(n,n)$ symmetry of $S(X)$ broken to $O(n,n;\mathbf{Z})$ "T-duality" of the superstring $\mathbf{S}(X)$.

## 1.2 T-theory

**Duff (1989) [7]:** String theories have a generalization that manifests T-duality at the cost of doubling the spacetime dimension $x \to (x, \tilde{x})$ for winding 0-modes.[1]

**Siegel (1993) [8–10]:** Doubled spacetime is reduced by a section condition on the momenta

$$(\partial_M) := \begin{pmatrix} \partial_\mu \\ \tilde{\partial}^\mu \end{pmatrix} \quad \eta := \begin{pmatrix} 0 & \delta^\nu_\mu \\ \delta^\mu_\nu & 0 \end{pmatrix} \quad \partial^M \partial_M = 0 \quad \Rightarrow \quad \text{e.g.} \quad \tilde{\partial}^m = 0.$$

Extension to a string theory $\mathbf{T}(X)$ with manifest T-duality symmetry.
New Lie derivative (D-bracket) and commutator (C-bracket) described by current algebra of generalized momenta $\Pi_M = P_M + \eta_{MN} X'^N$ with quantum bracket

$$[\Pi_M(\sigma_1), \Pi_N(\sigma_2)] = 2i\eta_{MN}\delta'(\sigma_1 - \sigma_2).$$

[1] Go Aggies!

Section condition arises dynamically as a new Virasoro constraint

$$L = \tfrac{1}{2}\eta^{MN}\Pi_M\Pi_N \quad \Rightarrow \quad \partial^M\partial_M = 0.$$

### 1.2.1 Generalized/Doubled Geometry

**Hitchin & Gualtieri (2003) [11, 12]:** Generalized geometry $T \oplus T^*$ as the massless $T(X)$ and further truncated (does not include winding 0-modes).
Dorfman bracket is generalized Lie derivative and Courant bracket is the generalized Lie bracket.

**Hull & Zwiebach (2009) [13]:** "Doubled geometries" $T(X)$ include coordinates for the winding modes. (Simplification of the manifestly T-dual "stringy geometry" $\mathbf{T}(X)$ of $X$ by Siegel.)

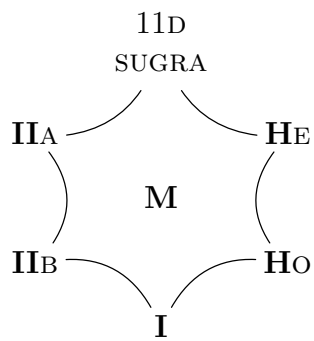
$$\begin{tikzcd}
 & \text{Doubled } T(X) \\
\text{Geometry } X \arrow[ur, dotted, "{[8\text{–}10,13]}"] \arrow[r, "{[11,12]}"'] & \text{Generalized} \arrow[u]
\end{tikzcd}
\qquad
\begin{tikzcd}
 & (T \oplus T^*)(X \times X) \\
X \arrow[ur, dotted] \arrow[r] & TX \oplus T^*X \arrow[u]
\end{tikzcd}$$

## 1.3 M-theory

**Cremmer, Julia, & Scherk (1978) [14]:** Construction of eleven-dimensional supergravity.

**Bergshoeff, Sezgin, & Townsend (1987) [15, 16]:** No strings in 11D but super-membranes.

**Witten (1995) [17]:** S-duality of the type-IIA **S**tring theory has 11-dimensional supergravity as its massless truncation.




**Cremmer & Julia (1979) [18] and Hull & Townsend [19]:** Analogously to the moduli space of compactifications of the type-II string theories on $n$-tori, hidden *exceptional* "U-duality" symmetry $E_{n(n)}$ of (gauged) 11D supergravity manifested by compactification on $T^n$.

| $n$ | $E_{n(n)}$ | $H_n$ | # vectors | # scalars |
|---|---|---|---|---|
| 3 | $SL(2;\mathbf{R}) \times SL(3;\mathbf{R})$ | $SO(2) \times SO(3)$ | 6 | 7 |
| 4 | $SL(5;\mathbf{R})$ | $SO(5)$ | 10 | 14 |
| 5 | $Spin(5,5)$ | $SO(5) \times SO(5)$ | 16 | 25 |
| 6 | $E_{6(6)}$ | $Sp(4)$ | 27 | 42 |
| 7 | $E_{7(7)}$ | $SU(8)$ | 28 | 70 |
| 8 | $E_{8(8)}$ | $SO(16)$ | 0 | 128 |

Table 1: Structure of compactifications of 11-dimensional supergravity on an $n$-torus.

**Duff (1985) [20] (see also [7, 21]):** "Either the $E_8 \times SO(16)$ structure appears only in $d = 3$ as an <u>internal</u> symmetry, or else it is really present at every $d \leq 11$ as a <u>spacetime</u> symmetry which transforms states of different spin into each other, in particular transforming [the frame field] into [the 3-form] in $d = 11$. We emphasize that we are advocating the latter."

## 1.4 F-theory

**Vafa (1996) [22]:** $SL(2;\mathbf{Z})$ S-duality of the type-IIB **S**tring may be realized geometrically in "12D" theory $F$ defined by

1. Geometrically realized (S(trong/weak) part of) U-duality
2. Not really 12D: Type-IIB **S**tring with axio-dilaton profile determined by elliptically-fibered Calabi-Yau 4-fold $T^2 \to \widehat{X} \to X$
3. Related by compactification to M-theory

$$\begin{tikzcd}
 & F(\widehat{X}) \arrow[dl, dotted] \arrow[dd] \\
M(X \times S^1) \arrow[dr] & \\
 & S(X)
\end{tikzcd}$$

### 1.4.1 Exceptional/Extended Geometry

**Berman & Perry (2010) [23]:** Extension of generalized geometry to "exceptional geometry"

$$\begin{tikzcd}
 & \text{Doubled } T(X) \arrow[r, "{[1–4]}"] & \text{F-theory } F(X) \\
\text{Geometry } S(X) \arrow[ur, dotted, "{[8–10,13]}"] \arrow[r, "{[11,12]}"'] & \text{Generalized} \arrow[u] \arrow[r, "{[23]}"'] & \text{Exceptional } M(X) \arrow[u]
\end{tikzcd}$$

corresponding to

$$\begin{tikzcd}
 & (T \oplus T^*)(X \times X) \arrow[r] & (T \oplus \Lambda^\bullet)(X \times X^\bullet) \\
X \arrow[ur, dotted] \arrow[r] & TX \oplus T^*X \arrow[u] \arrow[r] & TX \oplus \Lambda^\bullet X \arrow[u]
\end{tikzcd}$$

**Goal:** Provide a fundamental brane description of the ultra-violet completion of these theories analogous to fundamental strings $\mathbf{S}(X)$ for supergravity $S(X)$.

## 1.5 Definition of the fundamental theory $\mathbf{F}(X)$

### 1.5.1 F-gravity

**Definition of F-gravity $F(X)$:** We generalize the definition of F-theory by using both doubled and exceptional geometry: For any $X$, define $F(X)$ to make the following diagram commute ($n := D+1$)

$$\begin{tikzcd}
 & \begin{array}{c} F(X) \\ E_{n(n)}/H_{n(n)} \end{array} \arrow[dl, "\text{section}"'] \arrow[dr, "\text{reduction}"] & \\
\begin{array}{c} M(X) \\ \frac{GL(D+1,\mathbf{R})}{O(D,1)} \end{array} \arrow[dr, "\text{reduction}"'] & & \begin{array}{c} T(X) \\ \frac{O(D,D)}{O(D)\times O(D)} \end{array} \arrow[dl, "\text{section}"] \\
 & \begin{array}{c} S(X) \\ \frac{GL(D,\mathbf{R})}{O(D-1,1)} \end{array} &
\end{tikzcd}$$

### 1.5.2 Extension to $\mathbf{F}(X)$

**Goal: Extend to *fundamental* theories.** For each rank $n := D+1$ find brane theory (generalizing the string), the (first-)quantization of which gives the theory $\mathbf{F}(X)$ that completes the diagram

$$\begin{tikzcd}
 & \mathbf{F}(X) \arrow[dl, "\text{section}"'] \arrow[dr, "\text{reduction}"] & \\
\mathbf{M}(X) \arrow[dr] & & \mathbf{T}(X) \arrow[dl] \\
 & \mathbf{S}(X) &
\end{tikzcd}$$

It is *a priori* unclear whether this problem (existence of $\mathbf{F}$) has a solution. In fact, we have the following

**Result** ( [2–4]). *Let $D = \dim(X) \leq 6$. Then $\mathbf{F}(X)$ exists in the form of a current algebra with exceptional global symmetry $G = E_{n(n)}$ of rank $n = D+1$. This current algebra arises from the quantization of a free theory*

*with Hamiltonian symmetry $H_{n(n)}$ the (split form of the) maximal compact subgroup of $E_{n(n)}$. When a Lagrangian formulation is known ($D \leq 6$), the symmetry is enhanced to $L_{n(n)}$ and the theory is the Hamiltonian quantization of a free self-dual form on a brane of split signature.*

| $n$ | Lagrangian $L_{n(n)}$ | Hamiltonian $H_{n(n)}$ | Current Algebra $E_{n(n)}$ |
|---|---|---|---|
| 2 | $SL(2;\mathbf{R})$ | $SO(1,1)$ | $GL(2;\mathbf{R})$ |
| 3 | $SL(2;\mathbf{R})^2 \times SO(2)$ | $SL(2;\mathbf{R}) \times SO(2)$ | $SL(3;\mathbf{R}) \times SL(2;\mathbf{R})$ |
| 4 | $SL(4;\mathbf{R}) \simeq Spin(3,3)$ | $Sp(4;\mathbf{R}) \simeq Spin(3,2)$ | $SL(5;\mathbf{R})$ |
| 5 | $SL(4;\mathbf{C}) \simeq \mathrm{Spin}(6;\mathbf{C})$ | $Sp(4;\mathbf{C})$ | $Spin(5,5)$ |
| 6 | $Sp(8;\mathbf{C})$ | $USp(4,4)$ | $E_{6(6)}$ |
| 7 | $SU^*(8)$ | $SL(4;\mathbf{H})$ | $E_{7(7)}$ |

# 2 Case Study: $\mathbf{F}(X)$ for the 3D String

## 2.1 F-gravity

- Focus on case with $X$ modeled on $\mathbf{R}^{2,1}$.
- Supergravity $S(X)$ is 3D, $N = 2$ supergravity; Lorentz group is $SO(2,1)$.
- Manifestly T-dual 3D, $N = 2$ SG has T-duality group $SO(2,2) \sim SO(2,1)^2$.
- M-gravity $M(X)$ is 4D, $N = 1$ "old-minimal supergravity" (but with real prepotential for the conformal compensator) [1, 24].
- F-gravity $F(X)$ has global exceptional U-duality symmetry of rank 4: $G = E_{4(4)} \simeq SL(5;\mathbf{R})$.
- Coordinates $X^{mn}$ in the **10** (signature 4+6). But solution of strong constraint

$$\varepsilon^{mnpqr}\partial_{mn}\partial_{pq} = 0 \quad \Rightarrow \quad (\partial_{mn}) = (\partial_{-1\mu}, \partial_{\mu\nu}) \to (\partial_\mu, 0)$$

reduces this to (3+1)-dimensional $M(X)$ breaking $SL(5;\mathbf{R}) \to GL(4;\mathbf{R})$ spontaneously [25].

## 2.2 Virasoro algebra

**Strategy:** Attempt to interpret strong constraint as a Virasoro-type constraint for some worldvolume theory. In $\mathbf{T}(X)$ we saw the left-moving worldsheet current

$$\Pi_M(\sigma) = P_M + \eta_{MN} X'^N \quad \text{with} \quad [P_M(\sigma_1), X^N(\sigma_2)] = -i\delta_M^N \delta(\sigma_1 - \sigma_2)$$

and Virasoro constraint $L = \frac{1}{2}\eta^{MN}\Pi_M\Pi_N$ giving strong constraint $\partial^M\partial_M = 0$.
So we postulate for $\mathbf{F}(X)$ a wordvolume with 5 coordinates $\sigma^m$ (in addition to the $\tau$ coordinate associated to the Hamiltonian vector field) and $G = SL(5;\mathbf{R})$- covariant current

$$\Pi_{mn} = P_{mn} + \tfrac{1}{2}\varepsilon_{mnpqr}\partial^r X^{pq}.$$

Virasoro-type generator for strong constraint on string fields:

$$\mathcal{S}^r := \tfrac{1}{16}\varepsilon^{mnpqr}\Pi_{mn}\Pi_{pq}.$$

Implies $\varepsilon^{mnpqr}(\partial_{mn} f)(\partial_{pq} g) = 0$ for all functions $f$ and $g$ on $F(X)$.

But now algebra does not close:

$$[\mathcal{S}^m, \mathcal{S}^n] = \mathcal{S}^{(m}\,\partial^{n)}\delta + \partial^{[m}\mathcal{S}^{n]}\,\delta + \varepsilon^{mnpqr}\Pi_{pq}\mathcal{U}_r\,\delta \quad \text{with} \quad \mathcal{U}_r := \partial^s\Pi_{rs}.$$

Must interpret $\mathcal{U}$ as new constraint. It is central and reduces to

$$\mathcal{U}_m = \partial^n P_{mn}$$

... but this is a Gauß law constraint! Analogously to the first-quantization of Maxwell theory [26], first-quantize a 6D gauge 2-form $X_{MN}$:

| | gauge field | Lagrange multiplier | dynamical components | conjugate momentum | Gauß's law |
|---|---|---|---|---|---|
| Maxwell | $A_\mu$ | $A^\tau$ | $A^i$ | $E^i$ | $\partial^i E_i$ |
| $\mathbf{F}(X)$ | $X_{MN}$ | $X^{\tau m}$ | $X^{mn}$ | $P_{mn}$ | $\partial^n P_{mn}$ |

## 2.3 Chiral forms

**Siegel (1983) [27]:** By analogy with string action for chiral scalar fields, action for gauge 2-form with self-dual fieldstrength $F_{MNP}$:

$$S_{iegel} = \int \left( \tfrac{1}{12} F^{MNP} F_{MNP} + \tfrac{1}{4} \lambda^{MN} F_M^{(-)PQ} F_{NPQ}^{(-)} \right) d^6\sigma.$$

The equation of motion for $\lambda$

$$F_M^{(-)PQ} F_{NPQ}^{(-)} = 0 \quad \Rightarrow \quad F_{MNP}^{(-)} = 0$$

implies the fieldstrength of $X$ is self-dual whereas the gauge transformations

$$\delta X_{MN} = \alpha^P F_{MNP}^{(-)} + \tfrac{1}{12} \alpha^P \lambda_{[M}{}^Q F(-)_{NPQ]} + \mathcal{O}(\lambda^2)$$
$$\delta \lambda^{MN} = \partial^{(M} \alpha^{N)} + \alpha^P \partial_P \lambda^{MN} + \tfrac{1}{2} \lambda_P{}^{(M} \left( \partial^{N)} \alpha^P - \partial^P \alpha^{N)} \right) - \text{trace} + \mathcal{O}(\lambda^2)$$

imply $\lambda$ is pure gauge.

**Canonical analysis** In $\lambda \to 0$ gauge the Hamiltonian action becomes

$$S = -\int \tfrac{1}{2} \dot{X}^{mn} P_{mn} \, d^5\sigma d\tau + \int H \, d\tau$$
$$H = \int \left( \tfrac{1}{4} P^{mn} P_{mn} + \tfrac{1}{12} F^{mnp} F_{mnp} + X^{\tau m} \mathcal{U}_m \right) d^5\sigma.$$

Stress-energy tensor for selfdual field with $\Pi_{mn} := F_{\tau mn}^{(+)} = P_{mn} + \frac{1}{2} \varepsilon_{mnpqr} \partial^r X^{pq}$

$$\mathcal{T}_{mn} = \eta_{mn} \mathcal{T} - \tfrac{1}{2} \eta^{pq} \Pi_{mp} \Pi_{nq} \quad \boxed{\mathcal{S}^r = \tfrac{1}{16} \epsilon^{mnpqr} \Pi_{mn} \Pi_{pq}} \quad \mathcal{T} = \tfrac{1}{8} \eta^{mp} \eta^{nq} \Pi_{mn} \Pi_{pq}$$

Free quantum brackets $[P_{mn}(\sigma_1), X^{pq}(\sigma_2)] = -i \delta_{[m}^{[p} \delta_{n]}^{q]} \delta(\sigma_1 - \sigma_2)$ imply

$$\boxed{[\Pi_{mn}(\sigma_1), \Pi_{pq}(\sigma_2)] = 2i \epsilon_{mnpqr} \partial^r \delta(\sigma_1 - \sigma_2)}$$

guaranteeing enhanced symmetry: $E_{4(4)} = SL(5; \mathbf{R})$-covariant $\mathcal{SU}$ algebra above.

**Courant bracket** The algebra of spacetime gauge transformations generated by $\Lambda_i := \frac{i}{2} \int \lambda_i^{mn}(X(\sigma)) \Pi_{mn}(\sigma) \, d^5\sigma$

$$[\Lambda_1, \Lambda_2] = \Lambda_{12} \quad \text{with} \quad \lambda_{12}^{mn} = \tfrac{1}{4} \lambda_{[1}^{pq} \partial_{pq} \lambda_{2]}^{mn} - \tfrac{1}{2} \lambda_{[1}^{p[m} \partial_{pq} \lambda_{2]}^{n]q} - \tfrac{1}{4} \lambda_{[1}^{mn} \partial_{pq} \lambda_{2]}^{pq}$$

reproduces the "exceptional extended Courant bracket" of $E_{4(4)}$ computed by Berman, Godazgar,[2] and Perry [28].

**Conclusions:**

1. Quantization of the chiral 2-form on a 5-brane with split signature gives rise to a current algebra with $E_{4(4)}$ symmetry.

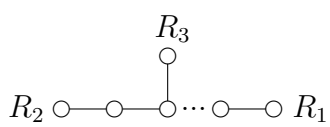
2. Solution of the constraints reduces the current algebra to that of the three-dimensional bosonic string.

3. The massless truncation of its C-bracket reproduces the "exceptional Courant bracket" of local symmetries of the M-gravity appropriate to this case.

# 3 Extensions

## 3.1 More U-duality

The U-duality $E_{4(4)}$ can be enlarged to $G = E_{n(n)}$.
Let $\mathbf{m} = 1, \ldots, \dim(R_1)$ where $R_1 = (0, \ldots, 1, 0)$ is the representation of $G$ corresponding to the left-most node in the Dynkin diagram arranged as




(This corresponds to $R_1 = \mathbf{10'}$, $R_2 = \mathbf{5}$, and $R_3 = \mathbf{5'}$ for $E_{4(4)} = SL(5;\mathbf{R})$). Similarly, we let $m$ run over $R_2 = (1, 0, \ldots, 0)$. ($R_3 = (0, \ldots, 0, 1)$ but in $SL(5;\mathbf{R})$ this is conjugate to $R_1$.)[2]

Introduce the $(R_1 \otimes R_1)_{sym} \supset R_2$ Clebsch-Gordan coefficients $\eta_{\mathbf{mn}p}$ and $\eta^{\mathbf{mn}p}$ and write the generalization of the current algebra

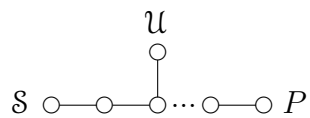
$$[\Pi_{\mathbf{m}}(\sigma_1), \Pi_{\mathbf{n}}(\sigma_2)] = 2i\eta_{\mathbf{mn}p}\partial^p\delta(\sigma_1, \sigma_2)$$

and Virasoro constraint

$$\mathcal{S}^p = \tfrac{1}{4}\eta^{\mathbf{mn}p}\Pi_{\mathbf{m}}\Pi_{\mathbf{n}}.$$

[2]These representations $R_p$ correspond to the number of $p$-forms in the supergravity tensor hierarchy of de Wit, Nicolai, and Samtleben [29].

Constraints in the diagram




Then the C-bracket of two vector fields is[3]

$$[V_1^{\mathbf{m}}\Pi_{\mathbf{m}}(\sigma_1), V_2^{\mathbf{n}}\Pi_{\mathbf{n}}(\sigma_1)] = 2i\eta_{\mathbf{mn}p}\partial^p\delta(\sigma_1-\sigma_2)V_1^{\mathbf{m}}V_2^{\mathbf{n}} \\ - i\delta(\sigma_1-\sigma_2)\left[V_{[1}^{\mathbf{m}}\partial_{\mathbf{m}}V_{2]}^{\mathbf{n}} - \tfrac{1}{2}\eta^{\mathbf{mn}r}\eta_{\mathbf{pq}r}V_1^{\mathbf{p}}\partial_{\mathbf{m}}V_2^{\mathbf{q}}\right]\Pi_{\mathbf{n}}.$$

Truncation to 0-modes should reproduce the $E_{n(n)}$ "Courant brackets" computed by Berman, Cederwall, Kleinschmidt, and Thompson [25]. Therefore, the projector $(R_1 \otimes R_1)_{sym} \to R_2$ is identified with the Clebch-Gordan coefficients

$$Y^{\mathbf{mn}}{}_{\mathbf{pq}} \leftrightarrow \eta^{\mathbf{mn}r}\eta_{\mathbf{pq}r}$$

up to constraints.
This $Y$ projector factorizes for all ranks $n \leq 6$ and factorizes up to a new constraint for $E_{7(7)}$ (corresponding to the 6D string).
Ultimate goal is 10-dimensional Lorentz invariance for $X$ corresponding to $E_{11}$ U-duality conjectured by West (2001) [30].

## 3.2 Fermions and critical superstrings

Critical superstrings can be constructed by adding in $10-D$ scalars $Y$ (and their duals $\tilde{Y}$) in the "ordinary string way". The Hamiltonian actions will have symmetry $H \times H'$ with $H' = SO(10-D)$.
Fermions are added for supersymmetry which still looks like

$$\{D_{\boldsymbol{\alpha}}(\sigma_1), D_{\boldsymbol{\beta}}(\sigma_2)\} = 2(\gamma^{\hat{\mathbf{m}}})_{\boldsymbol{\alpha\beta}}\Pi_{\hat{\mathbf{m}}}\delta(\sigma_1-\sigma_2)$$

but now the Clifford algebra is deformed to a Clifford-ish algebra

$$\gamma^{\hat{\mathbf{m}}}\gamma^{\hat{\mathbf{n}}} + \gamma^{\hat{\mathbf{n}}}\gamma^{\hat{\mathbf{m}}} = 2\eta^{\hat{\mathbf{m}}\hat{\mathbf{n}}}\mathbf{1} + 2\eta^{\hat{\mathbf{m}}\hat{\mathbf{n}}p}\gamma_p.$$

[3]Fields and operators on the right are evaluated at the midpoint $\frac{1}{2}(\sigma_1+\sigma_2)$.

Jacobi identities require the Fierz-ish identity

$$\eta_{\hat{\mathbf{m}}\hat{\mathbf{n}}p}(\gamma^{\hat{\mathbf{m}}})_{(\boldsymbol{\alpha}\boldsymbol{\beta}}(\gamma^{\hat{\mathbf{n}}})_{\boldsymbol{\gamma})\boldsymbol{\delta}} = 0$$

restricting $D + D' = 3, 4, 6, 10$.[4]

# 4 Retrospective

Why no quantizable membrane/higher-brane theories?
Either gapless [32–34] or no massless states at all [35, 36].
Novelties and possible advantages of this version of **F**:

1. Extended geometry of target $F(X)$ allows for more manifest U-duality.
2. Manifest U-duality grows as we increase the Lorentz invariance of $X$ (Duff's conjecture).
3. Reduction to the *stable* (critical super)string is *dynamical* instead of a by-hand truncation (same degrees of freedom as usual strings, just more 0-modes for "winding").
4. Extended geometry allows for conformal worldvolume theory: Instead of worldvolume scalar "immersion map" $\phi : \Sigma \to X$ with complicated dynamics, worldvolume field is a free self-dual $p$-form.

# 5 Open Questions

1. More U-duality? (Can we use Palmkvist [31] and related work to get $E_{>7}$?)
2. Relation to M2 and M5 in 11D target space of Bergshoeff, Sezgin, & Townsend (1987) [15, 16] and Howe & Sezgin (1996) [37, 38]?
3. Relation to gravitational tensor hierarchy?
4. Narian-type description in terms of exceptional Borcherds algebras?

[4] Also provides twistor solution $p_{\mathbf{m}} = \lambda\gamma_{\mathbf{m}}\lambda$ to the section condition $\mathcal{S} = 0$ modulo "pure spinors" $\lambda\gamma_{\mathbf{m}}\lambda = 0$.

5. Quantization?

6. Worldvolume gravity?

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